

Leveraging low-dimensional structure for efficient Gibbs posterior inference¹

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- Lots of interest recently on “generalized posteriors”
- e.g., a *Gibbs posterior distribution* is like Bayes but uses a loss function instead of likelihood⁴
- Well-known advantages:
 - little/no risk of model misspecification bias
 - direct attack on quantities of interest
- Wasserman writes:

The idea that statistical problems do not have to be solved as one coherent whole is anathema to Bayesians but is liberating for frequentists
- Generalized Bayes $\approx$ liberated Bayes (?)

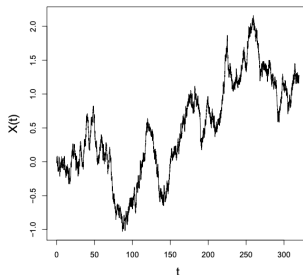
⁴Syring and M., arXiv:2012.04505 and arXiv:2203.09381

- Less-well-known properties:
 - extra flexibility, e.g., can reverse-engineer a suitable loss, say, when likelihood is intractable
 - can incorporate low-dim structural assumptions
- Focus here on a particular application:
 - discrete-time observations from a Lévy process
 - goal is (nonparametric) inference on the *Lévy density*
- This is an interesting & challenging problem, and similar results should be possible more generally

- Lévy process basics
- Gibbs posterior construction
 - loss function formulation
 - low-dim structure $\iff$ smoothness
- Results:
 - optimal (& adaptive) concentration rates
 - structure learning
- Illustration
- Concluding remarks

- Continuous-time process, $X = \{X(t) : t \in [0, T]\}$
 - *Independent increments*
 - *Stationary increments*, i.e., $X(t) - X(s) \stackrel{D}{=} X(t - s)$
 - *Stochastic continuity*, “ $P\{X \text{ has a jump at a fixed } t\} = 0$ ”
- Key property:⁵ “ $X = \text{continuous} + \text{jump}$ ”

- Examples:
 - Brownian motion
 - Compound Poisson
 - variance-gamma $\rightarrow$
- Interest is primarily in features of the jump part



⁵Lévy–Itô decomposition, requires mild conditions...

- Lévy process has three parameters: μ , σ , and Ψ
- Distribution of $X(t)$ most easily described by its char. fun

$$\gamma \mapsto \exp\left[t\left\{\mu i\gamma - \frac{\sigma^2\gamma^2}{2} + \int (e^{i\gamma x} - 1 - i\gamma x 1_{|x|<1}) \Psi(dx)\right\}\right]$$

- The *Lévy measure* Ψ controls the jump size & freq
- e.g., if X is compound Poisson, with intensity λ and jump size distribution J , then $\Psi = \lambda J$
- Assume Ψ has density ψ wrt Lebesgue measure
- My goal is probabilistic inference on ψ

- Two key challenges:
 - 1 Discrete-time observations $X(t_1), \dots, X(t_n)$
 - 2 difficult/impossible to write the likelihood for ψ
- Challenge 1 is subtle:
 - If we observed the whole path X , then we see the jumps and inference on Ψ relatively easy
 - Having only discrete samples means we can't distinguish jump parts from no-jump parts — everything looks like a jump!
- Challenge 2 is less-subtle: with no likelihood, a genuine Bayesian solution is out of reach

- Data structure:
 - set $Y_i = X(t_i) - X(t_{i-1})$, $i = 1, \dots, n$
 - sampling times $t_1, \dots, t_n$, with $t_i = i\Delta_n$ and $\Delta_n \rightarrow 0$
- Standard approach⁶ is to take a basis expansion $\psi = \psi_{\theta_K}$ indexed by K and $\theta_K = (\theta_{Kk} : k = 1, \dots, K)$
- For given K , there's a L_2 -based projection estimator $\psi_{\hat{\theta}_K}$ characterized by $\hat{\theta}_K = \arg \min_{\vartheta} R_{n,K}(\vartheta)$, where

$$R_{n,K}(\vartheta) = \|\vartheta\|^2 - \frac{2}{n\Delta_n} \sum_{i=1}^n \langle \vartheta, \underbrace{f_K(Y_i)}_{\text{basis}} \rangle, \quad \vartheta \in \Theta_K \subseteq \mathbb{R}^K$$

- This plays the role of *empirical expected loss*

⁶e.g., Figueroa-Lopez 2009; Comte & Genon-Catalot 2009

- Prior information about (K, θ_K) ?
- Low-dim structure (smoothness) encoded in prior for K
- Prior for ψ ...

$$\Pi(A) = \sum_K \underbrace{\tilde{\pi}(K)}_{\text{prior for } K} \underbrace{\tilde{\Pi}_K(\{\theta_K : \psi_{\theta_K} \in A\})}_{\text{cond prior for } \psi_{\theta_K}}$$

- Combine with empirical risk in a Bayesian-like way

$$\Pi_n(A) \propto \sum_K \tilde{\pi}(K) \int_{\{\theta_K : \psi_{\theta_K} \in A\}} \underbrace{e^{-\omega n \Delta_n R_{n,K}(\theta_K)}}_{\text{"likelihood"}} \tilde{\Pi}_K(d\theta_K)$$

- ω is a small *learning rate* parameter, important!

Asymptotic concentration rate

- Just a high-level summary here...
- Conditions:
 - 1 Prior for K : $\tilde{\pi}(K) \propto e^{-\beta K \log K}$, $\beta > 0$
 - 2 “High frequency sampling,” $n\Delta_n^{5/3} = O(1)$
 - 3 True ψ^* is Besov with smoothness α , *unknown*
 - 4 Learning rate ω is sufficiently small

Concentration rate theorem.

Under some conditions, the Gibbs posterior Π_n satisfies

$$\mathbb{E} \Pi_n(\{\psi : \|\psi - \psi^*\|_{L_2} > M_n \varepsilon_n\}) = o(1), \quad n \rightarrow \infty,$$

where $M_n \rightarrow \infty$ is arbitrary and

$$\varepsilon_n = \text{log-term} \times (n\Delta_n)^{-\alpha/(2\alpha+1)}.$$

Asymptotic concentration rate, cont.

■ Remarks:

- Rate is minimax optimal — even if path X is observed!
 - Adaptation to unknown smoothness of ψ^*
 - “ $n\Delta_n^{5/3} = O(1)$ ” can probably be improved
 - Sadly, “ ω sufficiently small” depends on a feature of ψ^*
- What about structure learning? Below is (to my knowledge) the first such result for Gibbs posteriors!

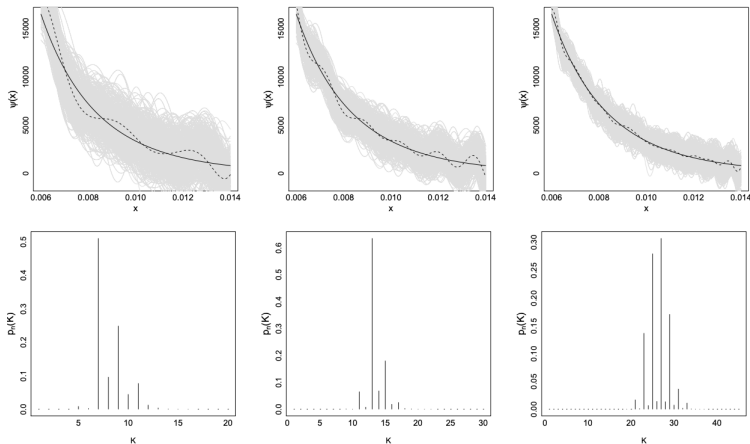
Structure learning theorem.

Under *basically* the same conditions, for some $\tau > 1$,

$$\mathbb{E} \tilde{\Pi}_n(\{K : K > \underbrace{\tau \lceil (n\Delta)^{1/(2\alpha+1)} \rceil}_{\text{“oracle” } K_n^*(\alpha)}\}) = o(1), \quad n \rightarrow \infty$$

- While a genuine Bayes solution to this problem is out of reach, the Gibbs solution is pretty straightforward
- Empirical risk is quadratic, $\approx$ Gaussian likelihood
- Conjugate normal prior for θ_K , given K
- Take prior for K as above
- Posterior computations:
 - marginal posterior for $K \rightarrow$ closed-form
 - conditional posterior for $(\theta_K | K) \rightarrow$ closed-form
- Note: nuisance parameters (μ, σ) are absent
- Only challenge is balancing the choice of (β, ω) ...

- Data from a *variance-gamma process*
- Three settings: consecutively more informative



- Incorporation of low-dim structure is important for efficiency
- Convenient to do so via a prior distribution, but Bayes doesn't have a monopoly on this
- Gibbs posteriors too can incorporate low-dim structure in a principled way and with good theoretical support
- Benefits include
 - no/little risk of model misspecification bias
 - direct attack on quantities of interest, i.e., no marginalization
- Should be possible to develop some general theory of structure learning for Gibbs posteriors...

Thanks for your attention!

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Gibbs posterior inference on a Lévy density under
discrete sampling

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